

Problem 2

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Problem 1 Suppose that $r(t) = r_0 e^{-kt}$ (with $k > 0$) is the rate at which a nation extracts oil. The current rate of extraction is $r(0) = 10^7$ barrels/yr. Also assume that the estimate of the total oil reserve (ie, the amount of oil remaining beneath the ground in this country) is 2×10^9 barrels.

- (a) Find $A(t)$, the total amount of oil extracted by the nation after t years.
- (b) Evaluate $\lim_{t \rightarrow \infty} A(t)$ and explain the meaning of this limit.
- (c) Find the minimum constant k for which the total oil reserves will last forever.
- (d) Suppose that the constant k is half the minimum value found in part (c). When will the nation deplete its oil reserve?